

ARK-493 → ARK-498 Series — Reconciliation Report

Series: ARK-493 through ARK-498

Preregistration: ARK-493-498-PREREGISTRATION-v1.1.md

Preregistration SHA-256: 464b9fb8be9d6cca052f236dc9deec9f8e89b781cafc58701e79b2d05d52952a —
verified match at execution time

Execution: single `python3 run_all.py run`, one shared hash-chained ProofRecord store

GATE-STOP triggered: No (`halted_by_gate_stop = false`)

This report reconciles what this series added to the standing ExecutionProof evidence corpus. Every number below is traceable to a file produced by this run (`results/results_ledger.jsonl` , `proofrecords/proofrecord_chain.jsonl` , `ledgers/T*.jsonl` , `results/execution_manifest.json` , `results/ark498_metrics.json`). No number is estimated.

1. This-series results (measured)

Experiment	Title	Decision	Scored cases	PASS	FAIL
ARK-493	Enforcement Boundary Under Adversarial Load	EXPERIMENT-PASS	90	90	0
ARK-494	Semantic Boundary — Deep Argument Mutation	EXPERIMENT-PASS	13	13	0
ARK-495	Temporal Boundary — Authority Change Mid-Flight	EXPERIMENT-PASS	11	11	0
ARK-496	Multi-Agent Delegation & Self-Approval Defense	EXPERIMENT-PASS	8	8	0
ARK-497	Independently Reconstructable ProofRecord	EXPERIMENT-PASS	30	30	0
ARK-498	Networked Production-Like Performance	EXPERIMENT-PASS	9	9	0
Total		6 / 6 PASS	161	161	0

- **Scored cases this series: 161** (preregistration minimum was 152, excluding ARK-498 per-request performance traffic). Met.
- **Dual-guard agreement: 161 / 161** scored cases (Guard-A in-process + Guard-B isolated subprocess agree on every record).
- **Enforcement leaks: 0.** No DENY or HOLD case produced an `executed` side-effect ledger entry in any tool ledger. The GATE-STOP rule (D-5) was therefore not triggered.
- **Failures preserved:** there were 0 failures to preserve this series; had any occurred, they would remain in the ledger with `failure_root_cause` and would not have been retried.

ARK-493 boundary matrix (the gate-stop gate)

- 6 execution paths × 15 cells = **90 cases**: P1-direct-call, P2-retry, P3-alternate-endpoint, P4-queued-execution, P5-tool-alias, P6-agent-created-subcall (15 each).
- Decision distribution: **30 ALLOW / 30 DENY / 30 HOLD**.
- All 30 ALLOW cases executed **exactly once** (idempotency held across retry/queued/alias paths).
All 60 DENY/HOLD cases executed **zero** times. **Leak rows: 0**.

Side-effect ledger totals (all scenarios, this run)

Invocation type	Count
executed	1,719
blocked (DENY)	139
held (HOLD)	30

(The large `executed` count is dominated by ARK-498's ~1,000 sustained-throughput ALLOW requests against T3; these are performance traffic, not scored cases.)

2. Corpus reconciliation

2.1 Pre-series baseline (as carried into this series)

Metric	Value
Experiment IDs	66
Case records	71
PASS	68
FAIL	2
GATE-STOP	1
Repositories	9

2.2 This series adds

Metric	Delta
Experiment IDs	+6 (ARK-493, -494, -495, -496, -497, -498)
Scored case records	+161
PASS	+161
FAIL	+0
GATE-STOP	+0
Repositories	+0 (added to the existing executionproof-testbeds repository as the ark-493-498/ subdirectory — no new repository created)

2.3 Projected post-series totals

Metric	Pre-series	+ This series	Post-series
Experiment IDs	66	+6	72
Case records	71	+161	232
PASS	68	+161	229
FAIL	2	+0	2
GATE-STOP	1	+0	1
Repositories	9	+0	9

Reconciled public headline numbers to use going forward: 72 experiments, 232 case records, 229 PASS, 2 FAIL, 1 GATE-STOP across 9 repositories.

The 2 pre-existing FAIL records and the 1 pre-existing GATE-STOP record are **retained unchanged**. Honest framing requires that the historical FAIL/GATE-STOP entries continue to be reported; they are evidence the methodology preserves negative results rather than suppressing them.

3. ARK-498 characterization (summary; full detail in ark498_performance_report.md)

PRODUCTION-LIKE OVERHEAD CHARACTERIZATION · NOT A BENCHMARK CERTIFICATION · NOT A PRODUCTION SLA

ARK-498 ran the gate behind a real Flask HTTP/loopback-TCP boundary across 9 scenarios (~1,810 requests) with simulated dependency latencies (policy lookup 50 ms, authority lookup 30 ms, real ProofRecord disk I/O).

- **Latency (measured, `time.monotonic()`):** cold start $\approx$ 85.2 ms; warm p50 $\approx$ 83.9 ms; under 10 concurrent clients p50 $\approx$ 84.4 ms / p95 $\approx$ 88.0 ms / p99 $\approx$ 93.9 ms.
- **Sustained throughput (middle 60% of a 1,000-request run, 5 clients):** $\approx$ 59.7 req/s. This figure is dominated by the injected 80 ms of simulated dependency sleeps per request and is **characterization data only**.
- **All six hard criteria met:** fail-closed leak count = 0 (S005/S006/S007); duplicate executions = 0 (S009); ProofRecord completeness = 100% (1,766/1,766); error accounting = 0 ALLOW-on-error; recovery with 0 auto re-executions of denied requests (S008); Guard-B signature verification = 100% (1,766/1,766).

These latency/throughput numbers **must not** be compared to the prior microsecond in-process test-bed (ARK-483–492): different architecture, different scope. They are not a production SLA and require independent measurement under production conditions before any performance claim.

4. Artifacts backing this report

Artifact	Path
Results ledger (161 scored rows)	<code>results/results_ledger.jsonl</code>
Hash-chained ProofRecords (1,894 records incl. summaries)	<code>proofrecords/proofrecord_chain.jsonl</code>
Side-effect ledgers T1–T5	<code>ledgers/T1_*.jsonl</code> ... <code>ledgers/T5_*.jsonl</code>
Execution manifest (prereg hash + per-experiment decisions)	<code>results/execution_manifest.json</code>
ARK-498 raw metrics	<code>results/ark498_metrics.json</code>
ARK-497 isolated-verifier output	<code>results/ark497_verifier_output.json</code>